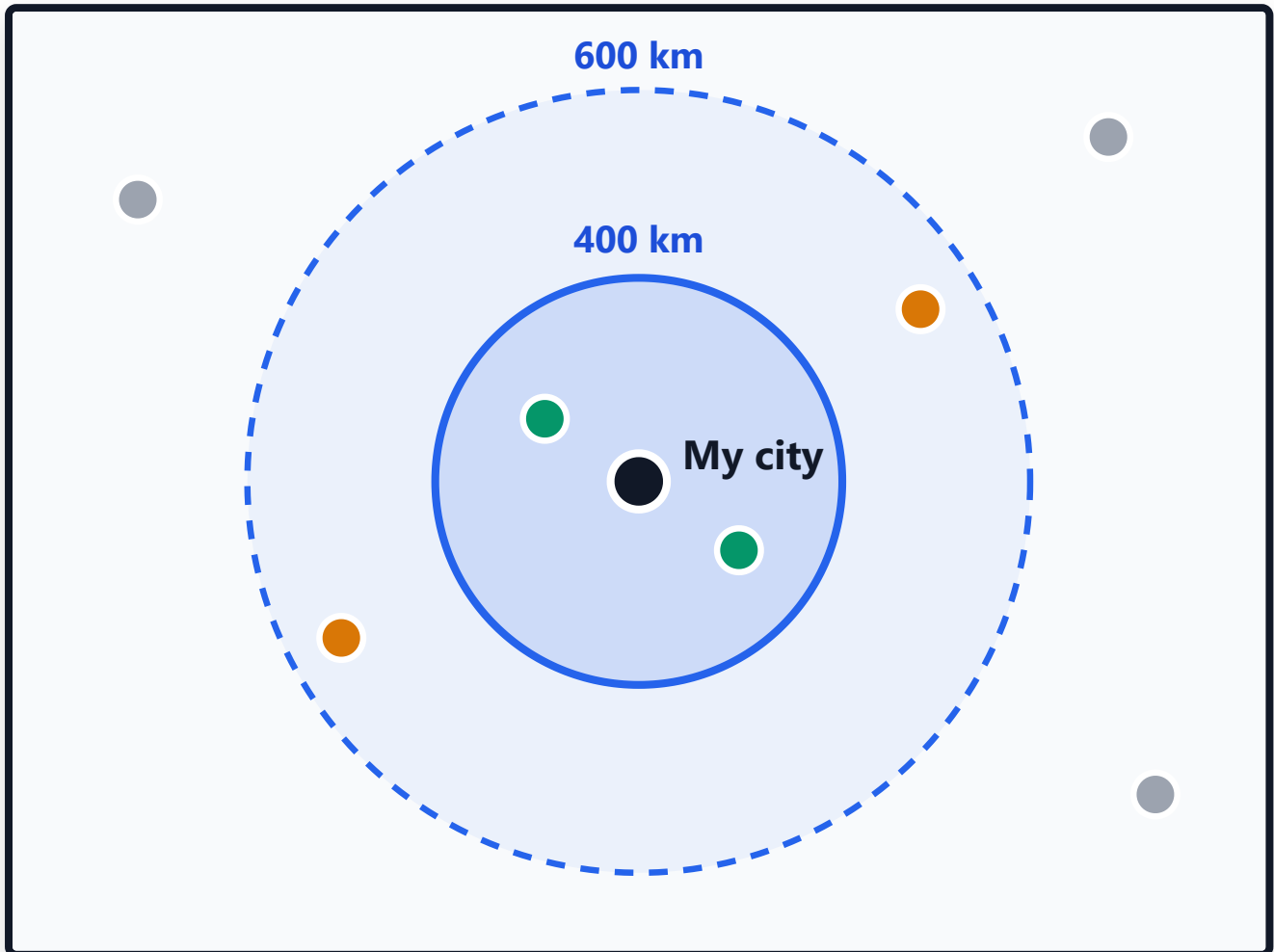


Near or Far?

Draw distance zones and ask your map a question, using GeoLibre



Which places are close to your city, and which are far away?

What you'll use

- **GeoLibre**: the same free, web-based GIS tool from Exercises 1 and 2
- **my_city.csv**: a one-row table that you will change to your own city
- **famous_places.csv**: the 16 famous places from Exercise 2
- **india_boundary.geojson**: the outline of India, from Exercise 1

Maps Can Answer Questions

So far you've used GIS to **draw** maps. But the real superpower of GIS is **asking questions** and getting answers from the map.

Today's question is: "**Which famous places are near my city?**" To answer it, you'll draw zones of a fixed distance around your city, then ask GeoLibre to find every famous place that falls inside them. Everyone in your class can pick a different city, so everyone gets a different answer!

GIS WORD OF THE DAY

Spatial analysis: using a map to answer a question about *where* things are, like "what is close to what?" or "what is inside what?"

FUN FACT

Real people use these tools every day. Ambulance services check which hospitals are closest, and city planners check which homes are near a new school or park.

Here's the plan for today, in seven steps:

1. Set up your map
2. Choose your city
3. Draw a 400 km zone around it
4. Make the zone see-through
5. Ask the map: which places are inside?
6. Count and compare with bigger zones
7. Make a layout and export your map

1

Set Up Your Map

Start with everything you already know how to do.

1. Open GeoLibre and click **Launch GeoLibre Web**.
2. Add **india_boundary.geojson** with **Add Data** → **Vector Layer**. Give it a transparent fill and a dark outline.
3. Add **famous_places.csv** with **Add Data** → **Delimited Text Layer**, using `longitude` and `latitude` for the coordinates.
4. Label the places with their `name`, so you can tell them apart.

TIP

Stuck? Exercise 1 shows how to add and style the India outline, and Exercise 2 shows how to load a CSV and add labels.

2

Choose Your City

Every analysis needs a starting point. Yours will be a city you choose.

1. Decide on your city. It could be where you live, where your grandparents live, or a city you'd love to visit.
2. In GeoLibre, find your city on the map, **right-click** on it, and copy the coordinate.
3. Open **my_city.csv** in Notepad. It looks like this:

```
name,latitude,longitude  
Bangalore,12.9716,77.5946
```
4. Change the second line to your city's name, latitude, and longitude. Save the file.
5. Load it with **Add Data** → **Delimited Text Layer**. Make its dot big and bold so it stands out from the famous places.

TIP

Watch the order! The copied coordinate may come as longitude first, then latitude. Latitudes in India are between about 8 and 37. Longitudes are between about 68 and 97. If your dot lands in the sea or another country, try swapping the two numbers.

3

Draw a 400 km Zone

Now let's draw a circle that covers everything within 400 km of your city.

1. Open **Processing** → **GeoLibre Toolbox** → **Vector**, and choose **Buffer**.
2. Set the input layer to *my_city*.
3. Set the distance to **400** kilometres. Check the unit carefully!
4. Click **Run**. A new layer appears: a big circle around your city.
5. Rename the new layer to something clear, like *zone_400km*.

GIS WORD OF THE DAY

Buffer: a zone drawn around something at a fixed distance. A buffer around a point is a circle. A buffer around a river would be a long, wiggly band.

FUN FACT

400 km is roughly a 7 to 8 hour car journey, or about an hour in a plane. Your zone shows everywhere you could reach "as the crow flies" with that much distance.

4

Make the Zone See-Through

Right now the zone may cover up the famous places underneath it. Let's fix that.

1. Select the *zone_400km* layer and open the **Style** panel.
2. Pick a fill colour you like, and turn the **fill opacity** down low, so the zone is see-through like tinted glass.
3. Give the zone a stronger **stroke** (outline) in the same colour, so its edge is easy to see.
4. In the Layers panel, drag the zone **below** *famous_places* and *my_city*, so the dots sit on top.

GIS WORD OF THE DAY

Opacity: how solid a colour is. 100% opacity is completely solid; 0% is completely invisible. Somewhere in between lets you see through it.

Ask the Map: Which Places Are Inside?

You could count the dots by eye, but GIS can do it for you, perfectly, every time.

1. Open **Processing** → **GeoLibre Toolbox** → **Vector**, and choose **Select by location**.
2. Set the input layer to *famous_places*. These are the things we want to find.
3. Set the overlay layer to *zone_400km*. This is the area we're searching in.
4. Set the relationship to **within**.
5. Click **Run**. The famous places inside your zone are now selected and highlighted on the map.

GIS WORD OF THE DAY

Spatial query: a question you ask a map using location, like "which places are *within* this zone?" It works just like a search, but for places.

TIP

If nothing is selected, that's an answer too! It means no famous places are within 400 km of your city. Bigger zones in the next step will catch some.

6

Count and Compare

Let's see how the answer changes as the zone grows.

1. Open the attribute table for *famous_places*. The status bar shows how many places are selected. Switch to **Show Selected** to see their names.
2. Write the number and the names in the table below.
3. Repeat Steps 3 and 5 with a **600 km** zone, and then a **1000 km** zone.

Zone size	How many places?	Which places?
400 km		
600 km		
1000 km		

My city: _____ Closest famous place: _____

FUN FACT

India measures about 3,200 km from north to south. Even a 1000 km zone covers only part of the country. How many 1000 km zones would it take to cover all of India?

7

Make a Layout and Export

Show off your answer with a finished map.

1. Turn on all three zones, with the biggest at the bottom of the Layers panel and the smallest just under the dots. Give each one a different colour.
2. Open **Project** → **Print Layout...**
3. Use a title like "**Famous Places Near [your city]**".
4. Turn on the **legend**, **north arrow**, and **scale bar**.
5. Export your map as a **PNG** or **PDF**.

TIP

Look at the scale bar on your exported map. Does the 400 km ring really look about 400 km from your city? This is a good way to check your work.

You Did Real Spatial Analysis!

You asked a question, used GIS tools to find the answer, and made a map to share it. That's exactly what professional GIS analysts do.

What You Learned

Using a map to answer questions with **spatial analysis**

Creating your own point from **coordinates**

Drawing distance zones with a **buffer**

Making layers see-through with **opacity**

Finding places inside a zone with a **spatial query**

Comparing results as the zone gets bigger

Try It Yourself

- Compare your table with a friend who picked a different city. Whose city has the most famous places nearby?
- Buffer **every** famous place by 100 km. Which zones overlap? Which place is the "loneliest", with no other place nearby?
- Try a tiny zone, like 20 km. Which places are practically next door to each other?
- Think of a real question you could answer with a buffer, like "which schools are within 2 km of my home?"